Implement Entra ID Integration with ABM for Account Auto Provisioning for Microsoft Intune

Implementation Solution Guide

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Executive Summary — Entra ID + Apple Business Manager Federation & Account Auto-Provisioning (Microsoft Intune)

Managed Apple IDs are the backbone of modern iPhone management — but hand-making them and reconciling them with your real users is a slog, and it blocks the cleanest enrollment experiences. The fix is to **integrate Microsoft Entra ID with Apple Business Manager**: **federate authentication** so users sign in to Apple services with their Entra credentials, and **auto-provision Managed Apple IDs** from Entra (SCIM) so accounts simply exist when you need them. With that foundation in place, all three iPhone models — **Corporate-Owned, BYOD, and Shared** — enroll into **Microsoft Intune** the right way.

This is a complete, **deploy-it-today** package, written for an administrator with no prior Apple-identity experience.

What you get

- A plain-language **Solution Overview** of Entra ↔ ABM federation, automatic Managed Apple ID provisioning (SCIM plus just-in-time on first sign-in), and the domain-capture / existing-Apple-ID conflict considerations.
- A **Requirements Guide** with verify / implement-if-missing for every prerequisite.
- An **Implementation Guide** of precise, click-by-click Entra and Intune steps (ABM actions named, exact ABM menus deferred to Apple's documentation).
- A dedicated **Enrollment Scenarios** section showing how the federated Managed Apple ID flows into **COD, BYOD, and Shared**, with the end-user sign-in experience per model.
- Five **PowerShell scripts** for readiness discovery, per-user/per-device validation, fleet coverage reporting, safe remediation, and scope-group building.
- **Testing** and **Troubleshooting** guides for confident rollout.

What this delivers

Federated authentication (Entra ID as the identity provider for Apple) and **self-provisioning Managed Apple IDs**, scoped to a pilot group first, with existing-Apple-ID conflicts resolved — then **Corporate-Owned (account-driven ADE)**, **BYOD (account-driven User Enrollment)**, and **Shared (no-user-affinity)** iPhones enrolling into Intune on one clean identity.

The deployment model

Assess → **Configure** → **Pilot** → **Validate** → **Expand** — federate and provision a scoped pilot, prove it, then expand.

Outcome

Entra ID federated with Apple Business Manager and Managed Apple IDs provisioning themselves — so every iPhone, whatever its purpose, enrolls into Intune the right way on one identity, validated before you

scale.

1. Solution Overview — Entra ID + Apple Business Manager Federation & Account Auto-Provisioning

What this solution is

This solution integrates **Microsoft Entra ID** with **Apple Business Manager (ABM)** so that two things happen automatically: users sign in to Apple services with their **Entra credentials** (federated authentication), and their **Managed Apple IDs** are created and kept in sync from Entra (automatic account provisioning). With that identity foundation in place, every company iPhone then enrolls into **Microsoft Intune** the right way for its purpose — **Corporate-Owned (COD)**, **BYOD**, or **Shared**.

It is the identity-and-provisioning layer beneath iPhone management. It does not replace your enrollment products; it makes them work on one clean identity.

The two pillars

Pillar 1 — Federated authentication (Entra ID is the identity provider for Apple)

You link Apple Business Manager to your Entra tenant and a **verified domain**. After federation, when a user signs in to an Apple service with their work email, Apple redirects the sign-in to Entra ID; the user authenticates with their normal Entra credentials (and MFA), and Apple trusts the result. There is no separate Apple password to manage.

Pillar 2 — Automatic Managed Apple ID provisioning

A **Managed Apple ID** is the organization-owned Apple account a user signs in with on managed Apple devices and services. This solution creates them automatically by **two complementary paths**:

1. **SCIM provisioning from Entra ID to Apple Business Manager.** Entra has a gallery

Apple Business Manager provisioning application. You scope it to a group, turn provisioning on, and Entra creates/updates the matching Managed Apple IDs in ABM on a recurring cycle — and de-provisions them when a user leaves the scope. This is the controlled, auditable bulk path.

1. **Just-in-time (JIT) creation on first federated sign-in.** With federation in place, a scoped

user who has no Managed Apple ID yet can have one created the first time they sign in to an Apple service with their federated credentials. This catches users who provisioning has not reached yet.

Use SCIM as the primary path for predictable, scoped, reversible provisioning; treat JIT as the backfill so no scoped user is ever blocked waiting for a cycle.

The domain-capture / existing-Apple-ID conflict (read this before you federate)

Federating a domain tells Apple that your organization owns every Apple identity that uses an email in that domain. If employees previously created **personal** Apple IDs using their work email (`name@yourdomain.com`), those personal accounts now **conflict** with the Managed Apple IDs your

organization wants to own — Apple calls this domain capture / a Managed Apple ID conflict. Apple gives those users a window to change their personal account to a different email so the company can claim the work-email identity. If this is ignored, provisioning fails for the conflicted users and their enrollment experience breaks. **Detecting and resolving these conflicts is a first-class step** in the Implementation Guide and a dedicated section in Troubleshooting — never skip the conflict scan before you widen the scope.

How the foundation feeds the three enrollment models

Once a user has a federated Managed Apple ID, Intune can enroll their iPhone the right way:

- **Corporate-Owned (COD)** — company-owned phones enrolled zero-touch via **Automated Device Enrollment (ADE) **from ABM,** supervised**, account-driven so the user signs in with their Entra identity / Managed Apple ID during setup.
- **BYOD** — personal phones enrolled with **account-driven User Enrollment**: a modern, privacy-preserving model that uses the **Managed Apple ID + Entra federated sign-in** to create a managed, cryptographically separated work partition while leaving the user's personal data and personal Apple ID untouched.
- **Shared** — appliance phones enrolled with **no user affinity** plus restrictions; staff identity lives in a **Managed Apple ID / shared app sign-in**, not bound to one person.

The detailed per-model flows and end-user experience are in **Section 4 — Enrollment Scenarios**.

An honest note on "Shared iPad" vs shared iPhone

Apple's true multi-user **Shared iPad** feature (where the device caches several users' Managed Apple IDs and each gets a partition) is an **iPad-only** capability. **iPhones do not have Shared iPad.** A "shared iPhone" in this solution means an iPhone enrolled with **no user affinity**, locked down with restrictions, where staff sign in to the relevant **app/service with a Managed Apple ID or a shared account** rather than personalizing the device. This solution states that accurately and builds the shared model on no-affinity enrollment, not on a feature iPhones lack.

What is in scope vs deferred

- **Exact navigation is given** for the **Entra side** (federation + the Apple Business Manager provisioning app) and the **Intune side** (APNs, ADE token, enrollment profiles, enrollment types).
- **Apple Business Manager console internals** (the exact ABM menu paths to add the federation / identity provider, manage Managed Apple IDs, and assign devices) change over time and are **deferred to Apple's official documentation** — this solution names the action and the control to use, and tells you what value to carry between consoles, without fabricating ABM menu paths.

Benefits

- **One identity** — users sign in to Apple with Entra credentials; no separate Apple password.
- **Hands-off accounts** — Managed Apple IDs provision and de-provision themselves with the user's

lifecycle (SCIM), with JIT as a safety net.

• **Right enrollment per phone** — COD, BYOD, and Shared each get the correct, modern model on the same federated identity.

• **Privacy for BYOD** — account-driven User Enrollment keeps personal data separate from work.

• **Auditable + reversible** — scoped to groups, piloted first, with scripts to validate and report.

Limitations (be honest with buyers)

• Federation is **tenant-and-domain-wide for Apple sign-in** — pilot on a scoped group and resolve existing-Apple-ID conflicts before you widen it; this is the highest-risk change in the solution.

• **ABM console specifics are deferred to Apple** (they move); this solution covers the Entra and Intune sides exactly and names the ABM actions.

• Provisioning from Entra covers **Managed Apple IDs in ABM**; it is not a general Apple data sync.

• Account-driven User Enrollment and account-driven device enrollment require **supported iOS versions** and a correctly federated Managed Apple ID — unsupported/old devices fall back to older enrollment flows.

The standard deployment model

> ***Federate ABM with Entra (scoped) → turn on Managed Apple ID provisioning for a pilot group → resolve existing-Apple-ID conflicts → stand up COD / BYOD / Shared enrollment → validate on the pilot → expand.*** *Pattern:* ***Assess → Configure → Pilot → Validate → Expand.***

2. Requirements Guide — Entra ID + ABM Federation & Account Auto-Provisioning

Verify **every** requirement before configuring. Each entry: **Description · Why required · How to verify · How to implement if missing.** Category-wide identity prerequisites are in `global-requirements-entra-id-solutions.md` (tenant, licensing, admin roles, break-glass accounts, UPN/domain hygiene, network) — confirm those first.

> ***Object names used throughout (use verbatim):*** *Entra provisioning app* `Apple Business Manager – Federation – Prod`*; provisioning job* `ABM SCIM Provisioning – Corp – Prod`*; provisioning pilot group* `Managed Apple ID Provisioning – Pilot – Stage`*; provisioning production group* `Managed Apple ID Provisioning – Corp – Prod`*; per-scenario groups* `iOS COD – Devices – Pilot`, `iOS BYOD – Users – Pilot`, `iOS Shared – Devices – Pilot` *(and …* `– Prod` *variants); ADE profiles* `iOS COD – ADE – Pilot`, `iOS Shared – ADE – Pilot`*; compliance* `iOS COD – Compliance – Pilot`, `iOS BYOD – Compliance – Pilot`, `iOS Shared – Compliance – Pilot`*.*

2.1 Microsoft Intune (and Entra ID P1+)

Description: Intune (Plan 1) for iPhone enrollment and policy; Entra ID P1+ for the federation and provisioning features (see global G2). **Why required:** Intune delivers the ADE / User Enrollment / no-affinity enrollment and compliance; Entra delivers federation and SCIM provisioning. **How to verify:** M365 admin center → **Billing** → **Licenses**; confirm Intune and Entra ID P1/P2. **How to implement if missing:** Assign M365 E3/E5 / Business Premium (Intune + Entra P1 included).

2.2 Apple Business Manager (ABM) account

Description: An organization **Apple Business Manager** account (not Apple School Manager) with an administrator who can configure identity/federation and manage devices and Managed Apple IDs. **Why required:** ABM is where federation is linked, Managed Apple IDs live, and devices are assigned to your MDM (Intune). **How to verify:** Sign in to `https://business.apple.com` as an ABM Administrator. **How to implement if missing:** Enroll the organization in Apple Business Manager (Apple docs); verify the organization and assign an ABM Administrator.

2.3 Verified domain(s) for federation

Description: The email domain(s) your Managed Apple IDs will use, **verified in Entra** and ready to be added in ABM for federation. **Why required:** Federation is keyed on a domain; Apple must confirm the org owns the domain, and the Managed Apple ID local part comes from the user's verified-domain email. **How to verify:** Entra → **Identity** → **Settings** → **Domain names** shows the domain **Verified**. **How to implement if missing:** Add and verify the public domain in Entra (DNS TXT record), then add it on the ABM side per Apple docs.

2.4 The Entra "Apple Business Manager" provisioning application

Description: The gallery **Apple Business Manager** enterprise application used to provision Managed Apple IDs by SCIM, added to your tenant as `Apple Business Manager – Federation – Prod`. **Why required:** It is the supported automatic provisioning path from Entra to ABM (scoped to groups, recurring sync, de-provision on removal). **How to verify:** Entra → **Identity** → **Applications** → **Enterprise applications**; confirm the app exists; run `ABMPROV-Discovery.ps1`. **How to implement if missing:** Add it from the gallery (Implementation Step 2) and name it `Apple Business Manager – Federation – Prod`.

2.5 Apple MDM Push (APNs) certificate

Description: A valid **Apple MDM Push certificate** connecting Intune to Apple for iOS management. **Why required:** No iPhone can be managed by Intune without an active APNs certificate; it underpins all three enrollment models. **How to verify:** Intune → **Devices** → **Enrollment** → **Apple** → **Apple MDM Push certificate**; status **Active**; run `ABMPROV-Discovery.ps1` (reports expiry). **How to implement if missing:** Create/renew it (Implementation Step 6) with a **company** Apple ID you record and reuse yearly.

2.6 ADE enrollment program token (for Corporate-Owned)

Description: An **Automated Device Enrollment** token linking an ABM **MDM server** entry to Intune, so company-owned devices enroll zero-touch. **Why required:** COD enrollment is driven by the ADE token; devices assigned to the Intune MDM server in ABM flow to Intune and receive an ADE profile. **How to verify:** Intune → **Devices** → **Enrollment** → **Apple** → **Enrollment program tokens**; a valid, non-expired

token exists; run `ABMPROV-Discovery.ps1`. **How to implement if missing:** Create it (Implementation Step 7): download Intune's public key, create the MDM server in ABM, upload the token back to Intune.

2.7 Supported iOS versions for modern enrollment

Description: iPhones on an iOS version that supports **account-driven User Enrollment** (BYOD) and **account-driven device enrollment** (COD). **Why required:** The modern, Managed-Apple-ID-based flows require recent iOS; older devices fall back to legacy enrollment. **How to verify:** Check device iOS versions in Intune → **Devices** → **All devices**; confirm against Apple's stated minimums for account-driven enrollment. **How to implement if missing:** Update target iPhones to a supported iOS version before enrolling.

2.8 User/group scoping and a pilot group

Description: Entra security groups that scope provisioning and each enrollment model, plus a small **pilot** group, all created with the naming convention above. **Why required:** Federation and provisioning must be **piloted on a scoped group first** — you do not federate/provision the whole tenant at once. Per-scenario groups keep COD, BYOD, and Shared cleanly separated. **How to verify:** Entra → **Groups**; confirm the provisioning + per-scenario groups exist; run `ABMPROV-Report.ps1`. **How to implement if missing:** Create them with `ABMPROV-Set-ScopeTargeting.ps1` (Implementation Step 1).

2.9 Existing-Apple-ID conflict review readiness

Description: A plan and access to detect personal Apple IDs that use your verified-domain email (domain capture conflicts) and resolve them before widening federation. **Why required:** Conflicted users fail Managed Apple ID provisioning and cannot complete account- driven enrollment until resolved. **How to verify:** ABM provides the conflict / domain-capture report on the federation/identity screen (deferred to Apple docs); plan user communications for the change window. **How to implement if missing:** Run the conflict review in ABM (Implementation Step 4), notify affected users to change their personal Apple ID email, then re-check before expanding.

2.10 Enrollment-time grouping owner for the Shared device group (Intune Provisioning Client as Owner)

Description: For the **Shared** (no-affinity) scenario, when `iOS Shared – Devices – Pilot` is an **Assigned** security group (the deterministic targeting model), the **Intune Provisioning Client** first-party service principal — **AppId** `f1346770-5b25-470b-88bd-d5744ab7952c`, often shown as `Intune Autopilot ConfidentialClient` — must be an **Owner** of that group. **Why required:** Enrollment-time device grouping adds each newly enrolled shared iPhone to the Assigned group by way of this principal *owning* the group. Without the owner the group stays **empty at enrollment** and **none of the shared-device compliance/config profiles or apps deploy**. This is the same first-party principal and mechanism Windows Autopilot device preparation uses. Applies only when a **security group targets** the shared devices; a Dynamic group instead relies on its membership rule and does not use this owner. **How to verify:** Entra → **Groups** → `iOS Shared – Devices – Pilot` → **Owners** lists the service principal — confirm by **AppId** `f1346770-5b25-470b-88bd-d5744ab7952c`, not the display name (which varies by tenant). Verify in your tenant before each pilot. **How to implement if missing:** Entra → **Groups** → `iOS Shared – Devices – Pilot` → **Owners** → **Add owners**, search by the AppId, and add it. (If the

principal isn't present, open the Intune portal once as Global Administrator to provision it, then add it as owner.)

Checklist

- Intune + Entra ID P1+ licensing assigned (2.1)
- Apple Business Manager account with an Administrator (2.2)
- Verified domain(s) for federation (2.3)
- Entra `Apple Business Manager – Federation – Prod` provisioning app added (2.4)
- Apple MDM Push (APNs) certificate Active (2.5)
- ADE enrollment program token valid (2.6) — for Corporate-Owned
- Target iPhones on a supported iOS version (2.7)
- Provisioning + per-scenario scope groups created, incl. pilot (2.8)
- Existing-Apple-ID conflict review planned (2.9)
- Intune Provisioning Client (AppId `f1346770-5b25-470b-88bd-d5744ab7952c`) is an Owner of the Assigned `iOS Shared – Devices – Pilot` group (2.10)

3. Implementation Guide — Entra ID + ABM Federation & Account Auto-Provisioning

Follow these steps **in order**. The Entra side and the Intune side are documented click-by-click; the Apple Business Manager side names the action and defers the exact ABM menu paths to Apple's official documentation (they change over time).

Consoles used:

- **Microsoft Entra admin center** — `https://entra.microsoft.com`
- **Microsoft Intune admin center** — `https://intune.microsoft.com`
- **Apple Business Manager (ABM)** — `https://business.apple.com` (menu paths deferred to Apple docs)

> ***Model:*** *Federate ABM with Entra, turn on Managed Apple ID provisioning* ***scoped to a pilot group****, resolve existing-Apple-ID conflicts, then stand up COD / BYOD / Shared enrollment, validate on the pilot, and expand.* ***Assess*** *→* ***Configure*** *→* ***Pilot*** *→* ***Validate*** *→* ***Expand.***
>
> ***Artifact names (use verbatim):*** *provisioning app* `Apple Business Manager – Federation – Prod`*; provisioning job* `ABM SCIM Provisioning – Corp – Prod`*; provisioning pilot group* `Managed Apple ID Provisioning – Pilot – Stage`*; provisioning prod group* `Managed Apple ID Provisioning – Corp – Prod`*; per-scenario groups* `iOS COD – Devices – Pilot`*,* `iOS BYOD – Users – Pilot`*,* `iOS Shared – Devices – Pilot`*; ADE profiles* `iOS COD – ADE – Pilot`*,* `iOS Shared – ADE – Pilot`*.*

> ***Risk + rollback (read first):*** *Federating a domain changes Apple sign-in for* ***everyone*** *whose email is in that domain. Always* ***scope provisioning to the pilot group first*** *and resolve conflicts before widening.* ***Rollback:*** *set the provisioning job to* ***Off*** *and remove the pilot group's assignment to stop creating Managed Apple IDs; the federation link itself is removed on the ABM side (deferred to Apple docs) only if you must fully revert — coordinate this, as it affects sign-in.*

Phase 1 — Scope groups and the provisioning application

Step 1 — Create the scope groups

1. Entra → **Groups** → **New group** → **Security** → `Managed Apple ID Provisioning – Pilot – Stage` (a few

representative users). **Create**.

1. Repeat for `Managed Apple ID Provisioning – Corp – Prod` (the full population, used later at expand).

2. Create the per-scenario groups: `iOS COD – Devices – Pilot`, `iOS BYOD – Users – Pilot`, and

`iOS Shared – Devices – Pilot`. **Create** each.

> *Prefer automation? Run* `ABMPROV-Set-ScopeTargeting.ps1` *to create all of these groups in one pass (default* `-WhatIf`*; add* `-Apply`*).*

Step 2 — Add the Apple Business Manager provisioning application (Entra)

1. Entra → **Identity** → **Applications** → **Enterprise applications** → **New application**.

2. Search the gallery for **Apple Business Manager**, select it, and add it. **Name it**

`Apple Business Manager – Federation – Prod`. **Create**.

1. Open `Apple Business Manager – Federation – Prod` → **Users and groups** → **Add user/group** → add

`Managed Apple ID Provisioning – Pilot – Stage` only (pilot scope). **Assign**.

Phase 2 — Federate ABM with Entra ID

Step 3 — Link Apple Business Manager to Entra ID (federated authentication)

1. In **ABM**, open the identity / federation settings and **add Microsoft Entra ID as the identity

provider**, then** federate your verified domain** (the exact ABM screen/menu is deferred to Apple's official documentation).

1. Complete the trust between ABM and Entra: when ABM prompts, **sign in / consent with an Entra

administrator** so Apple can redirect sign-ins for that domain to Entra ID.

1. The value that links the two sides is your **verified domain** (Requirement 2.3) — confirm it is the

domain you intend to federate before saving. After this, users in that domain authenticate to Apple services via Entra.

> *Federation now covers the **whole domain** for Apple sign-in. Provisioning, however, is still scoped to the pilot group from Step 2 — keep it that way until conflicts are resolved (Step 4) and the pilot passes.*

Step 4 — Detect and resolve existing-Apple-ID conflicts (do this before widening)

1. In **ABM**, open the **domain capture / Managed Apple ID conflict** report for the federated domain

(the exact ABM location is deferred to Apple docs). It lists personal Apple IDs that use an email in your domain.

1. For each conflicted user, Apple provides a process to have the user **change their personal Apple ID

to a different (personal) email**, freeing the work-email identity for a Managed Apple ID. Notify those users with a deadline (the change window).

1. Re-open the conflict report and confirm it is **clear for the pilot users** before turning on

provisioning. Do not expand beyond the pilot while conflicts remain for in-scope users.

Phase 3 — Automatic Managed Apple ID provisioning

Step 5 — Configure SCIM provisioning of Managed Apple IDs (Entra side, pilot-scoped)

1. Entra → **Identity** → **Applications** → **Enterprise applications** → **open**

`Apple Business Manager – Federation – Prod` → **Provisioning** → **Get started**.

1. Set **Provisioning Mode** to **Automatic**.

2. Under **Admin Credentials**, authorize Entra to ABM as the connector requires (for the Apple

Business Manager gallery app this is an OAuth authorization to ABM; complete the ABM-side authorization prompt — deferred to Apple docs for the exact ABM screen). **Test Connection**; when it succeeds, treat this provisioning configuration as `ABM SCIM Provisioning – Corp – Prod`.

1. Under **Mappings**, confirm the user attribute mappings ABM expects — at minimum a stable user

identifier and the attributes that form the **Managed Apple ID** (the verified-domain email / UPN).

1. Under **Settings**, set **Scope** to **Sync only assigned users and groups**, then **Save**. (Only

`Managed Apple ID Provisioning – Pilot – Stage` is assigned, so only the pilot provisions.)

Step 6 — Start provisioning to the pilot scope

1. On the **Provisioning** page, set **Provisioning Status** to **On**.

2. Provisioning runs on a recurring cycle; the first cycle creates Managed Apple IDs in ABM for the

members of `Managed Apple ID Provisioning – Pilot – Stage`. Monitor **Provisioning logs** for success and address any per-user errors (often a remaining conflict from Step 4).

1. Run `ABMPROV-Discovery.ps1` (provisioning job status) and `ABMPROV-Validation.ps1`

(per-user Managed Apple ID provisioned/federated) for the pilot users.

> ***Just-in-time backfill:*** *with federation in place, a scoped user without a Managed Apple ID can have one created on* ***first federated sign-in*** *to an Apple service. Use SCIM as the primary path and JIT as the safety net so no pilot user is blocked waiting for a cycle.*

Phase 4 — Intune Apple connection (shared by all enrollment models)

Step 7 — Create the Apple MDM Push (APNs) certificate

1. Intune → **Devices** → **Enrollment** → **Apple** → **Apple MDM Push certificate**.
2. Check **I agree**, **Download** the CSR.
3. **Create your MDM push certificate** → `https://identity.apple.com`; sign in with a **company Apple

ID** (record it — reuse it every year).

1. **Create a Certificate**, upload the CSR, **Download** the `.pem`; back in Intune enter the **Apple

ID**, Browse **to the** `.pem`**,** Upload**. Confirm status** Active** (~1 year out).

Step 8 — Connect Apple Business Manager to Intune (ADE token — for Corporate-Owned)

1. Intune → **Devices** → **Enrollment** → **Apple** → **Enrollment program tokens** → **Add** → check **I agree** →

Download the Intune **public key** (`.pem`).

1. In **ABM**, add an **MDM server** and **upload the Intune public key**, then **download the server

token** (`.p7m`) (exact ABM menu deferred to Apple docs). Name the MDM server `Intune – iOS COD`.

1. Back in Intune, enter the **Apple ID** used for ABM, **Browse** to the `.p7m`, **Upload**. The token

appears under **Enrollment program tokens**.

Phase 5 — Stand up the three enrollment models

Step 9 — Corporate-Owned (COD): ADE enrollment profile (supervised, account-driven)

1. Intune → **Devices** → **Enrollment** → **Apple** → **Enrollment program tokens** → open the token from Step 8

→ **Profiles** → **Create profile** → **iOS/iPadOS**. **Name it** `iOS COD – ADE – Pilot`; add a description.

1. **User affinity** = **Enroll with user affinity**, and select **account-driven** enrollment so the

user signs in with their **Entra identity / Managed Apple ID** during Setup Assistant.

1. **Management options:** enable **Supervised**; set **Locked enrollment = Yes**.
2. **Setup Assistant:** hide non-essential panes for a clean first-run.
3. **Review + Create.**
4. Assign devices: in **ABM**, assign the company-owned iPhone serials to the `Intune – iOS COD`

MDM server (deferred to Apple docs); in Intune, on the token → **Devices**, **Sync** (Apple rate-limits — allow a few minutes), select the serials → **Assign profile** → `iOS COD – ADE – Pilot`.

Step 10 — BYOD: account-driven User Enrollment

1. Intune → **Devices** → **Enrollment** → **Enrollment types** (Apple) → **Create** a profile for

User Enrollment; **name it** `iOS BYOD – User Enrollment – Pilot`.

1. Set the enrollment type to **account-driven User Enrollment** (the modern, privacy-preserving model

that uses the **Managed Apple ID + Entra federated sign-in**).

1. **Assignments:** assign to `iOS BYOD – Users – Pilot` (users, not devices — BYOD is user-scoped).

Create.

> *No ADE token is needed for BYOD; the user starts enrollment from the device's settings and authenticates with their federated Managed Apple ID, which creates the managed work partition.*

Step 11 — Shared: no-user-affinity enrollment

1. On the ADE token, **Profiles** → **Create profile** → **iOS/iPadOS**; **name it** `iOS Shared – ADE – Pilot`.

2. **User affinity** = **Enroll without user affinity** (this is what makes the device shared — no

primary user). Enable **Supervised**; **Locked enrollment = Yes**.

1. **Setup Assistant:** hide non-essential panes. **Review + Create.**

2. In **ABM**, assign the shared iPhone serials to the `Intune – iOS COD` MDM server (or a dedicated

server entry) (deferred to Apple docs); in Intune, on the token → **Devices** → **Sync**, select the serials → **Assign profile** → `iOS Shared – ADE – Pilot`.

1. Staff identity on shared phones lives in the **Managed Apple ID / shared app sign-in**, not bound to

a primary user. (iPhones have no Apple "Shared iPad" — see Overview.)

> ***Shared device group targeting — make*** `iOS Shared – Devices – Pilot` ***an Assigned group owned by the Intune Provisioning Client (enrollment-time grouping).*** *For deterministic shared-device enrollment, keep* `iOS Shared – Devices – Pilot` *(created in Step 1) an* ***Assigned*** *security group and add the* ***Intune Provisioning Client*** *first-party service principal —* ***AppId*** `f1346770-5b25-470b-88bd-d5744ab7952c`*, in many tenants displayed as* `Intune Autopilot ConfidentialClient` *— as an* ***Owner*** *of the group. In Entra:* ***Groups*** → `iOS Shared – Devices – Pilot` → ***Owners*** → ***Add owners****, and search by that* ***AppId*** *(match by AppId,* ***not*** *the display name, which varies by tenant). Enrollment-time device grouping works by this principal *owning* the Assigned group — without the owner the group stays* ***empty at enrollment*** *and* ***none of the shared-device profiles (e.g.*** `iOS Shared – Compliance – Pilot`***) or apps deploy****. The group must be* ***Assigned****, never Dynamic, for this mechanism. This is the same first-party principal and mechanism Windows Autopilot device preparation uses.* ***Verify the owner is present before each pilot.*** **(If the principal isn't present in the tenant, open the Intune portal once as Global Administrator to provision it, then add it as owner.)**

Step 12 — Compliance policies per model

1. Intune → **Devices** → **Compliance** → **Create policy** → **iOS/iPadOS**; create one each:

`iOS COD – Compliance – Pilot`, `iOS BYOD – Compliance – Pilot`, `iOS Shared – Compliance – Pilot`.

1. Set rules appropriate to each model (e.g. min iOS version; jailbreak detection; a passcode rule for

COD/BYOD; relaxed/no passcode for hand-off-speed shared phones).

1. **Assignments:** `iOS COD – Compliance – Pilot` → `iOS COD – Devices – Pilot`;

`iOS BYOD – Compliance – Pilot` → `iOS BYOD – Users – Pilot`; `iOS Shared – Compliance – Pilot` → `iOS Shared – Devices – Pilot`. **Create** each.

Phase 6 — Pilot, validate, expand

Step 13 — Pilot-enroll one device per model

1. **COD:** erase a pilot company iPhone (**Settings → General → Transfer or Reset → Erase All Content

and Settings**), power on, and complete** Remote Management** — it enrolls supervised, account-driven, and the user signs in with their federated Managed Apple ID.

1. **BYOD:** on a personal iPhone, start **account-driven User Enrollment** from device settings and

sign in with the federated Managed Apple ID — a managed work partition is created.

1. **Shared:** erase a pilot shared iPhone and enroll via ADE with `iOS Shared – ADE – Pilot` — it

enrolls **no user affinity**; staff sign in to the relevant app/service with the Managed Apple ID.

Step 14 — Validate the pilot

1. Run `ABMPROV-Discovery.ps1` (federation + provisioning job + ADE token + APNs readiness).

2. Run `ABMPROV-Validation.ps1` (per-user Managed Apple ID provisioned/federated; per-device correct

enrollment model).

1. Run `ABMPROV-Report.ps1` (provisioning coverage + COD/BYOD/Shared breakdown). Complete the

Testing Guide pilot gate before expanding.

Step 15 — Expand to production

1. **Provisioning:** Entra → `Apple Business Manager – Federation – Prod` → **Users and groups** → add

`Managed Apple ID Provisioning – Corp – Prod`; the next cycle provisions the full population. Confirm conflicts are clear org-wide first.

1. **COD / Shared:** add remaining serials to the ADE token and assign the … `– ADE – Pilot` profiles

(or make them the **default** profile for the token); add devices to the per-scenario groups.

1. **BYOD:** extend `iOS BYOD – User Enrollment – Pilot` to the broader BYOD user population.

2. Re-run `ABMPROV-Report.ps1` and roll out in waves, validating each wave before widening.

Order of operations

`Scope groups → ABM provisioning app (Entra) → federate ABM↔Entra → resolve conflicts → SCIM provisioning (pilot) → APNs → ADE token → COD ADE / BYOD User Enrollment / Shared no-affinity → compliance → pilot-enroll → validate → expand`

4. Enrollment Scenarios — How the Federated Identity + Managed Apple ID Flows into Each Model

This section details how the same foundation — **Entra federation + an auto-provisioned Managed Apple ID** — feeds each of the three iPhone models, and what the **end-user sign-in experience** looks like per model. All three are first-class; keep them cleanly separated by the per-scenario groups from the Implementation Guide.

> ***Shared identity, three behaviors.*** *In every model the user authenticates to Apple via **Entra ID** and uses a **Managed Apple ID** provisioned from Entra. What differs is *user affinity* (who the device belongs to), *supervision*, and *where the identity is bound* (to the device, to the user's work partition, or to an app/service).*

4.1 Corporate-Owned (COD) — supervised, account-driven device enrollment

What it is: A company-owned iPhone, enrolled zero-touch via **Automated Device Enrollment (ADE)** from Apple Business Manager, **supervised**, with **user affinity** so the device is tied to its assigned employee.

How the identity flows:

- The device serial is assigned to the `Intune – iOS COD` MDM server in ABM and receives the `iOS COD – ADE – Pilot` profile.
- During **Setup Assistant**, the phone shows **Remote Management** for your organization. Because the profile is **account-driven**, the user signs in with their **Entra credentials**; their **auto-provisioned Managed Apple ID** is used for managed Apple services.
- Device sits in `iOS COD – Devices – Pilot` and gets `iOS COD – Compliance – Pilot`.

End-user sign-in experience: Power on a new/erased phone → connect to network → see "Remote Management" naming the company → **sign in with work email + Entra password (and MFA)** → the phone finishes setup already managed and supervised, signed in with the Managed Apple ID. No personal Apple ID, no separate Apple password.

Use when: the organization owns the phone and wants full control and zero-touch provisioning.

4.2 BYOD — account-driven User Enrollment (privacy-preserving)

What it is: A personal iPhone enrolled with **account-driven User Enrollment**, the modern model that uses the **Managed Apple ID + Entra federated sign-in** to create a **cryptographically separated managed (work) partition**, leaving the user's personal data and **personal Apple ID** untouched.

How the identity flows:

- The user is a member of `iOS BYOD – Users – Pilot` (user-scoped, not device-scoped).
- The user starts enrollment from **device settings**; iOS prompts for the **work account**, redirects

the sign-in to **Entra** (federated), and uses the **federated Managed Apple ID** to stand up the managed work partition.

- The device receives `iOS BYOD – Compliance – Pilot`; only work apps/data live in the managed

partition — Intune cannot see or wipe personal data.

End-user sign-in experience: Settings → add the work account / start enrollment → **sign in with work email + Entra password (and MFA)** → a separate, managed work area appears for company apps; the personal side (personal Apple ID, photos, messages) is unchanged. The user keeps their own Apple ID for personal use **and** has a Managed Apple ID for work.

Use when: employees use personal phones and privacy/separation matters; no ADE token is required.

4.3 Shared — no user affinity + Managed Apple ID / shared app sign-in

What it is: An appliance iPhone enrolled via ADE with **no user affinity**, supervised, locked down. Staff are **not** bound to the device as primary users; identity lives in the **Managed Apple ID / a shared sign-in to the relevant app/service**.

Accurate scope: iPhones do **not** have Apple's multi-user **Shared iPad** feature (that is iPad-only). "Shared iPhone" here means **no-affinity enrollment + restrictions**, with staff signing in to the app/service that matters using a Managed Apple ID or a shared account — not personalizing the device.

How the identity flows:

- The device serial is assigned to its MDM server in ABM and receives `iOS Shared – ADE – Pilot`

(**Enroll without user affinity**, **Supervised**, **Locked enrollment**).

- The device joins `iOS Shared – Devices – Pilot` and gets `iOS Shared – Compliance – Pilot`.
- Staff identity is **not** the device login; it is the **Managed Apple ID / shared app sign-in** used

inside the relevant managed app or service.

> ***Targeting note (Assigned group + owner):*** *When* `iOS Shared – Devices – Pilot` *is an* ***Assigned*** *security group, enrollment-time device grouping only populates it if the* ***Intune Provisioning Client*** *service principal (****AppId*** `f1346770-5b25-470b-88bd-d5744ab7952c`*, often shown as* `Intune Autopilot ConfidentialClient`*) is an* ***Owner*** *of the group — otherwise the group stays empty at enrollment and no shared-device profiles/apps deploy. Match by* ***AppId****, not the display name, and verify in your tenant. See Implementation Step 11 and Requirement 2.10. The group must be* ***Assigned****, never Dynamic, for this mechanism (the same principal Windows Autopilot device preparation uses).*

End-user sign-in experience: The phone is already set up and locked down — no per-user device sign-in. A staff member picks it up and **signs in to the relevant app/service** (with a Managed Apple ID or a shared account) to do their task, then hands it on. The device itself stays generic.

Use when: phones are shared across staff/shifts (front desk, floor, kiosk-style) and you want a generic, controlled device rather than a personal one.

4.4 Keeping the three cleanly separated

- **COD** → device group `iOS COD – Devices – Pilot`, profile `iOS COD – ADE – Pilot`.
- **BYOD** → user group `iOS BYOD – Users – Pilot`, profile `iOS BYOD – User Enrollment – Pilot`.
- **Shared** → device group `iOS Shared – Devices – Pilot`, profile `iOS Shared – ADE – Pilot`.

Never assign more than one model's enrollment profile / compliance policy to the same device or user. Use `ABMPROV-Set-ScopeTargeting.ps1` to build the groups consistently and `ABMPROV-Report.ps1` to confirm each device lands in exactly one model.

Model	Owner	User affinity	Supervised	Identity is bound to	Needs ADE token
COD	Company	Yes	Yes	The device (assigned user)	Yes
BYOD	Employee	User-scoped	No	The managed work partition	No
Shared	Company	No (no affinity)	Yes	The app/service (Managed Apple ID)	Yes

5. Testing Guide — Entra ID + ABM Federation & Account Auto-Provisioning

Run after provisioning is on for the pilot and one device per model is enrolled (Implementation Steps 13–14), and again after each expansion wave. Use `ABMPROV-Discovery.ps1` / `ABMPROV-Validation.ps1` / `ABMPROV-Report.ps1` to back each test with data.

Test 1 — Federation is active for the domain

Do: Have a pilot user sign in to an Apple service with their work email; confirm the sign-in redirects to **Entra** and accepts their Entra credentials (and MFA). **Expected:** Authentication completes via Entra ID — no separate Apple password prompt.

Test 2 — Existing-Apple-ID conflicts are clear for pilot users

Do: Re-check the ABM domain-capture / conflict report for the pilot users (deferred to Apple docs). **Expected:** No conflicted personal Apple IDs remain for in-scope pilot users.

Test 3 — Managed Apple IDs auto-provision (SCIM)

Do: Entra → `Apple Business Manager – Federation – Prod` → **Provisioning** → **Provisioning logs**; run `ABMPROV-Validation.ps1` for the pilot users. **Expected:** Pilot users have a **Managed Apple ID provisioned** (provisioning job healthy; logs show success for the pilot group).

Test 4 — Just-in-time backfill works

Do: With a scoped user who has no Managed Apple ID yet, have them sign in to an Apple service with their federated credentials. **Expected:** A Managed Apple ID is created on first federated sign-in (JIT) without a manual step.

Test 5 — COD enrolls correctly

Do: Run `ABMPROV-Validation.ps1` for a pilot COD device; check Intune → **Devices**. **Expected:** The phone is **supervised**, **has user affinity**, enrolled via **ADE** (`iOS COD – ADE – Pilot`), in `iOS COD – Devices – Pilot`, compliant.

Test 6 — BYOD enrolls correctly (privacy-preserving)

Do: Run `ABMPROV-Validation.ps1` for a pilot BYOD device; check Intune → **Devices**. **Expected:** The phone shows **account-driven User Enrollment**, a **managed work partition**, user in `iOS BYOD – Users – Pilot`; personal data is not managed.

Test 7 — Shared enrolls correctly (no affinity)

Do: Run `ABMPROV-Validation.ps1` for a pilot shared device; check Intune → **Devices**. **Expected:** The phone is **supervised**, **no primary user**, enrolled via `iOS Shared – ADE – Pilot`, in `iOS Shared – Devices – Pilot`; staff sign in to the app/service with a Managed Apple ID.

Test 8 — Apple connection readiness

Do: Run `ABMPROV-Discovery.ps1`. **Expected:** APNs certificate **Active** (not near expiry), a valid **ADE token**, the provisioning job **On/healthy**, and federation reachable.

Test 9 — Fleet coverage + scenario breakdown

Do: Run `ABMPROV-Report.ps1`. **Expected:** A clear provisioning-coverage figure and a COD / BYOD / Shared device breakdown; each device appears in exactly one model.

Pass/fail gate

Gate	Requirement
Federation/provisioning pass	Tests 1–4 pass for the pilot group

Gate	Requirement
Per-model pass	Tests 5–7 pass (one device validated per model)
Readiness pass	Tests 8–9 pass before any expansion wave

6. Troubleshooting Guide — Entra ID + ABM Federation & Account Auto-Provisioning

Use `ABMPROV-Discovery.ps1` for federation/provisioning/ADE/APNs readiness, `ABMPROV-Validation.ps1` for per-user/per-device state, `ABMPROV-Report.ps1` for fleet coverage, and `ABMPROV-Remediation.ps1` (default `-WhatIf`) for safe fixes. ABM-side console specifics are deferred to Apple's official documentation.

A. Federation conflicts / existing personal Apple IDs

Symptom	Cause	Remediation
User can't get a Managed Apple ID; "in use"	A personal Apple ID already uses the work-email (domain capture conflict)	Re-run the ABM domain-capture report; have the user change their personal Apple ID to a different email; re-check before re-provisioning
Federated sign-in fails for some users	Those users still have a conflicting personal Apple ID, or were not in the change window	Resolve the conflict (above); confirm the user is in the federated, provisioned scope
Whole-domain sign-in concerns after federation	Federation is domain-wide for Apple sign-in	Expected — keep provisioning scoped to the pilot group until conflicts clear; communicate the change

B. SCIM provisioning failures (Entra → ABM)

Symptom	Cause	Remediation
Provisioning job won't start / Test Connection fails	ABM authorization not completed or expired	Re-authorize ABM under Provisioning → Admin Credentials; Test Connection (ABM screen deferred to Apple docs)
Some users never provision	They aren't in the assigned scope group, or "Sync only assigned" excludes them	Add them to `Managed Apple ID Provisioning – Pilot – Stage/– Corp – Prod`; run `ABMPROV-Remediation.ps1` to fix membership

Symptom	Cause	Remediation
Per-user errors in Provisioning logs	Attribute mapping / unresolved conflict	Check Mappings (stable identifier + verified-domain email); resolve any Apple-ID conflict; let the next cycle retry
Quota / not-found errors	App not assigned, or scope set to "all" by mistake	Confirm the app is assigned to the scope group and Scope = "Sync only assigned users and groups"

C. COD (Corporate-Owned) enrollment issues

Symptom	Cause	Remediation
Device doesn't show "Remote Management"	Serial not assigned to the Intune MDM server in ABM, or token not synced	Assign the serial to `Intune – iOS COD` in ABM (deferred to Apple docs); Intune → token → Devices → Sync (allow for Apple rate limits)
Enrolls but not supervised / not account-driven	Wrong ADE profile or profile not assigned	Confirm `iOS COD – ADE – Pilot` has Supervised + account-driven and is assigned to the serial
User can't sign in during setup	Federation/Managed Apple ID not ready for that user	Confirm the user is federated and provisioned (`ABMPROV-Validation.ps1`) before enrolling their device

D. BYOD (account-driven User Enrollment) issues

Symptom	Cause	Remediation
Enrollment offers the wrong (older) flow	iOS too old for account-driven User Enrollment, or wrong enrollment type	Update iOS (Requirement 2.7); confirm the profile is account-driven User Enrollment assigned to `iOS BYOD – Users – Pilot`
Work account won't add	Federated Managed Apple ID missing for the user	Provision/JIT the Managed Apple ID first; verify with `ABMPROV-Validation.ps1`
Personal data appears managed	Device was enrolled with device enrollment, not User Enrollment	Re-enroll via account-driven User Enrollment (the privacy-preserving model)

E. Shared (no-affinity) enrollment issues

Symptom	Cause	Remediation
Device asks for a primary user	Profile has user affinity	Use `iOS Shared – ADE – Pilot` with **Enroll without user affinity**
Expecting Apple "Shared iPad" on iPhone	iPhones have no Shared iPad feature	Use no-affinity + restrictions + Managed Apple ID / shared app sign-in (see Overview / Section 4)
Staff can't access the app	Shared sign-in not set up	Sign in to the relevant app/service with the Managed Apple ID or shared account
`iOS Shared – Devices – Pilot` is **empty after enrollment**; no shared-device profiles or apps deploy	The group is **Assigned** but the **Intune Provisioning Client** SP (**AppId** `f1346770-5b25-470b-88bd-d5744ab7952c`, often shown as `Intune Autopilot ConfidentialClient`) is **not an Owner**, so enrollment-time device grouping can't populate it	Entra → **Groups** → `iOS Shared – Devices – Pilot` → **Owners** → **Add owners**, search by that **AppId** (not the display name), then re-enroll a pilot device. Group must be **Assigned**, not Dynamic (same principal Windows Autopilot device preparation uses). If the SP is missing, open the Intune portal once as Global Administrator to provision it, then add it as owner

F. Apple connection (APNs / ADE token)

Symptom	Cause	Remediation
No iPhones can be managed	APNs certificate expired or wrong Apple ID	Renew APNs with the **same** company Apple ID (Implementation Step 7); never switch the Apple ID
COD devices stop flowing from ABM	ADE token expired	Renew the enrollment program token (re-upload the new `.p7m`) — `ABMPROV-Discovery.ps1` reports expiry

G. Diagnostics quick reference

- `ABMPROV-Discovery.ps1` — federation app, provisioning job status, ADE token, APNs readiness.
- `ABMPROV-Validation.ps1` — per-user Managed Apple ID + per-device enrollment model.
- `ABMPROV-Report.ps1` — provisioning coverage + COD/BYOD/Shared breakdown.

• Entra → `Apple Business Manager – Federation – Prod` → **Provisioning** → **Provisioning logs** for

per-user provisioning results.

H. When to escalate

If federation is configured, conflicts are clear, the provisioning job is On and healthy, the user is in the assigned scope, and a Managed Apple ID still won't provision after a cycle, capture the Provisioning log entry and the ABM domain-capture report state and open a Microsoft (Entra provisioning) and/or Apple (ABM) support case as the failure side indicates.

7. Best Practices — Entra ID + ABM Federation & Account Auto-Provisioning

• **Pilot the scope, not the federation alone.** Federation is domain-wide for Apple sign-in, so the

control you have is provisioning scope — keep SCIM scoped to `Managed Apple ID Provisioning – Pilot – Stage` until conflicts are resolved and the pilot passes. This is the single most important practice.

• **Resolve existing-Apple-ID conflicts before widening.** Run the ABM domain-capture report, give

users a clear change-window deadline, and re-check before expanding. Conflicts are the number-one cause of provisioning and enrollment failures.

• **Use SCIM as primary, JIT as backfill.** SCIM gives scoped, auditable, reversible provisioning and

de-provisioning with the user lifecycle; just-in-time creation on first federated sign-in catches anyone a cycle hasn't reached. Rely on both.

• **Keep the three models cleanly separated.** One enrollment profile and one compliance policy per

device/user; COD by device group, BYOD by user group, Shared by device group. Never double-assign.

• **Be accurate about shared iPhones.** iPhones have no Apple "Shared iPad" — build shared on

no-affinity enrollment + restrictions + a Managed Apple ID / shared app sign-in, and say so.

• **Name everything consistently.** Use the `<Purpose> – <Scope> – <Stage>` convention verbatim across

groups, ADE profiles, and compliance policies so docs, scripts, and the console all match.

• **Protect the Apple credentials.** Record the **company Apple ID** used for the APNs certificate and

the **ADE token** Apple ID; both renew yearly — a lapse breaks all iPhone management.

• **Keep break-glass ready.** Per the global requirements, keep cloud-only emergency-access accounts

excluded from broad policies before changing identity/sign-in.

• **Validate every wave.** Run `ABMPROV-Discovery.ps1`, `ABMPROV-Validation.ps1`, and

`ABMPROV-Report.ps1` before and after each expansion; don't widen on assumption.

• **Have a rollback.** To stop provisioning, set the job **Off** and remove the pilot group's

assignment; full federation removal is an ABM-side action (deferred to Apple docs) — coordinate it because it affects sign-in.

- **De-provision with the lifecycle.** Let SCIM remove Managed Apple IDs when a user leaves scope; do

not orphan accounts.

8. Deliverables Summary — Entra ID + ABM Federation & Account Auto-Provisioning

Documentation

Document	Purpose
Executive Summary	Business case and what's in the box
1. Solution Overview	Federation + Managed Apple ID provisioning (SCIM + JIT), the conflict nuance, and how it feeds COD/BYOD/Shared
2. Requirements Guide	Every prerequisite with Verify / Implement-if-missing
Global Requirements (Entra ID Solutions)	Category-wide identity prerequisites, published once
3. Implementation Guide	Exact, ordered Entra + Intune steps; ABM actions named, menus deferred to Apple
4. Enrollment Scenarios	How the federated Managed Apple ID flows into COD, BYOD, and Shared, with the per-model sign-in experience
5. Testing Guide	Validation procedures with pass/fail gates
6. Troubleshooting Guide	Federation/conflict, SCIM, and per-scenario enrollment issues
7. Best Practices	Field-proven recommendations

PowerShell scripts

Script	What it does
`ABMPROV-Discovery.ps1`	Readiness: federation app, Entra→ABM provisioning job status, ADE token, APNs (read-only)
`ABMPROV-Validation.ps1`	Per-user Managed Apple ID provisioned/federated; per-device correct enrollment model (read-only)
`ABMPROV-Report.ps1`	Fleet provisioning coverage + COD/BYOD/Shared scenario breakdown (read-only)
`ABMPROV-Remediation.ps1`	Safe fixes (default `-WhatIf`): re-provision a user / fix scope-group membership

Script	What it does
`ABMPROV-Set-ScopeTargeting.ps1`	Build the provisioning + per-scenario enrollment scope groups (default `-WhatIf`)

What the buyer receives

- This implementation guide as a branded **Word** document and **PDF**.
- All five **PowerShell scripts** in a single ZIP.
- A repeatable process: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand.**

Outcome

Entra ID federated with Apple Business Manager and Managed Apple IDs provisioning themselves — so Corporate-Owned, BYOD, and Shared iPhones all enroll into Intune cleanly on one identity, proven on a pilot before you scale.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

ABMPROV-Discovery.ps1

Purpose: Reports the Entra "Apple Business Manager" provisioning application, its SCIM synchronization job status (Managed Apple ID provisioning), the Intune Apple MDM Push (APNs) certificate, and the Automated Device Enrollment (ADE) enrollment program token(s) — so you can assess readiness before federating and enrolling. Read-only against Microsoft Graph. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\ABMPROV-Discovery.ps1 Requires the Microsoft Graph PowerShell SDK and delegated scopes: Application.Read.All, Synchronization.Read.All, DeviceManagementServiceConfig.Read.All, DeviceManagementConfiguration.Read.All, Group.Read.All. EXPECTED RESULTS The provisioning app + object id, the SCIM job status, the APNs certificate state/expiry, and each ADE token with its expiry. ABM-side federation/identity-provider configuration is deferred to Apple's official documentation. .NOTES Entra ID Solutions — Entra ID + ABM federation & auto-provisioning. Read-only. ABM console deferred. > [CmdletBinding()] param([string]$AppDisplayName = 'Apple Business Manager - Federation - Prod', [string]$PilotGroup = 'Managed Apple ID Provisioning - Pilot - Stage')

How to run: powershell -ExecutionPolicy Bypass -File .\ABMPROV-Discovery.ps1 Requires the Microsoft Graph PowerShell SDK and delegated scopes: Application.Read.All, Synchronization.Read.All, DeviceManagementServiceConfig.Read.All, DeviceManagementConfiguration.Read.All, Group.Read.All. EXPECTED RESULTS The provisioning app + object id, the SCIM job status, the APNs certificate state/expiry, and each ADE token with its expiry. ABM-side federation/identity-provider configuration is deferred to Apple's official documentation. .NOTES Entra ID Solutions — Entra ID + ABM federation & auto-provisioning. Read-only. ABM console deferred. > [CmdletBinding()] param([string]$AppDisplayName = 'Apple Business Manager - Federation - Prod', [string]$PilotGroup = 'Managed Apple ID Provisioning - Pilot - Stage')

ABMPROV-Remediation.ps1

Purpose: Closes the most common provisioning gaps: (1) add a user to the Managed Apple ID provisioning scope group so the next SCIM cycle provisions them; (2) ensure the provisioning scope group is assigned to the Entra "Apple Business Manager" application; and (3) optionally request an on-demand provisioning retry for a user (provisionOnDemand). By default it runs in -WhatIf mode and only reports the changes it would make; pass -Apply to make them. It does NOT configure federation or resolve Apple-ID conflicts — those are ABM-side and deferred to Apple's docs. HOW TO RUN Preview (default): powershell -ExecutionPolicy Bypass -File .\ABMPROV-Remediation.ps1 -UserUpn user1@contoso.com Apply: powershell -ExecutionPolicy Bypass -File .\ABMPROV-Remediation.ps1 -UserUpn user1@contoso.com -Apply Requires the Microsoft Graph PowerShell SDK and delegated scopes: Application.ReadWrite.All, AppRoleAssignment.ReadWrite.All, Group.ReadWrite.All, User.Read.All, Synchronization.ReadWrite.All. EXPECTED RESULTS In preview: the membership/assignment/retry actions that are needed. With -Apply: the user is added to the scope group, the group is assigned to the app if missing, and (if -OnDemand) a provisioning retry is requested. Re-run ABMPROV-Validation.ps1 / ABMPROV-Report.ps1 afterward. .NOTES Entra ID Solutions — Entra ID + ABM federation & auto-provisioning. Default -WhatIf. ABM/conflicts deferred. > [CmdletBinding(SupportsShouldProcess=$true)] param([Parameter(Mandatory=$true)][string]$UserUpn, [string]$AppDisplayName = 'Apple Business Manager - Federation - Prod', [string]$ScopeGroup = 'Managed Apple ID Provisioning - Pilot - Stage',

[switch]$OnDemand, [switch]$Apply)

How to run: Preview (default): powershell -ExecutionPolicy Bypass -File .\ABMPROV-Remediation.ps1 -UserUpn user1@contoso.com Apply: powershell -ExecutionPolicy Bypass -File .\ABMPROV-Remediation.ps1 -UserUpn user1@contoso.com -Apply Requires the Microsoft Graph PowerShell SDK and delegated scopes: Application.ReadWrite.All, AppRoleAssignment.ReadWrite.All, Group.ReadWrite.All, User.Read.All, Synchronization.ReadWrite.All. EXPECTED RESULTS In preview: the membership/assignment/retry actions that are needed. With -Apply: the user is added to the scope group, the group is assigned to the app if missing, and (if -OnDemand) a provisioning retry is requested. Re-run ABMPROV-Validation.ps1 / ABMPROV-Report.ps1 afterward. .NOTES Entra ID Solutions — Entra ID + ABM federation & auto-provisioning. Default -WhatIf. ABM/conflicts deferred. > [CmdletBinding(SupportsShouldProcess=$true)] param([Parameter(Mandatory=$true)][string]$UserUpn, [string]$AppDisplayName = 'Apple Business Manager - Federation - Prod', [string]$ScopeGroup = 'Managed Apple ID Provisioning - Pilot - Stage', [switch]$OnDemand, [switch]$Apply)

ABMPROV-Report.ps1

Purpose: Produces a fleet report: provisioning scope coverage (how many users are in the provisioning scope groups and whether the SCIM job is active), plus a breakdown of managed iOS devices by enrollment model — Corporate-Owned (COD), BYOD, and Shared — so you can audit that every iPhone enrolled the intended way on the federated identity. Read-only against Microsoft Graph. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\ABMPROV-Report.ps1 Requires the Microsoft Graph PowerShell SDK and delegated scopes: Application.Read.All, Synchronization.Read.All, Group.Read.All, DeviceManagementManagedDevices.Read.All. EXPECTED RESULTS Provisioning coverage (scope-group member counts + job status) and a COD / BYOD / Shared device count breakdown with a per-device list. ABM-side provisioned-account state is deferred to Apple. .NOTES Entra ID Solutions — Entra ID + ABM federation & auto-provisioning. Read-only. ABM state deferred. > [CmdletBinding()] param([string]$AppDisplayName = 'Apple Business Manager - Federation - Prod', [string[]]$ScopeGroups = @('Managed Apple ID Provisioning - Pilot - Stage','Managed Apple ID Provisioning - Corp - Prod'))

How to run: powershell -ExecutionPolicy Bypass -File .\ABMPROV-Report.ps1 Requires the Microsoft Graph PowerShell SDK and delegated scopes: Application.Read.All, Synchronization.Read.All, Group.Read.All, DeviceManagementManagedDevices.Read.All. EXPECTED RESULTS Provisioning coverage (scope-group member counts + job status) and a COD / BYOD / Shared device count breakdown with a per-device list. ABM-side provisioned-account state is deferred to Apple. .NOTES Entra ID Solutions — Entra ID + ABM federation & auto-provisioning. Read-only. ABM state deferred. > [CmdletBinding()] param([string]$AppDisplayName = 'Apple Business Manager - Federation - Prod', [string[]]$ScopeGroups = @('Managed Apple ID Provisioning - Pilot - Stage','Managed Apple ID Provisioning - Corp - Prod'))

ABMPROV-Set-ScopeTargeting.ps1

Purpose: Ensures the Entra security groups this solution relies on exist, using the solution's naming convention: the Managed Apple ID provisioning scope groups (pilot + prod) and the per-scenario enrollment groups for Corporate-Owned, BYOD, and Shared. Optionally adds members by UPN. Creates the groups only; assigning the provisioning group to the Apple Business Manager app is done by ABMPROV-Remediation.ps1, and federation is configured in ABM (deferred to Apple docs). Default -WhatIf; pass -Apply to make changes. HOW TO RUN Preview (default): powershell -ExecutionPolicy Bypass -File .\ABMPROV-Set-ScopeTargeting.ps1 Apply: powershell -ExecutionPolicy Bypass -File

.\ABMPROV-Set-ScopeTargeting.ps1 -Apply Add pilot members: ... -Apply -ProvisioningPilotMembers user1@contoso.com,user2@contoso.com Requires the Microsoft Graph PowerShell SDK and delegated scopes: Group.ReadWrite.All, User.Read.All. EXPECTED RESULTS The provisioning (pilot/prod) and per-scenario (COD devices, BYOD users, Shared devices) security groups exist (created if missing); any supplied members are added. Re-run ABMPROV-Report.ps1 to confirm membership. .NOTES Entra ID Solutions — Entra ID + ABM federation & auto-provisioning. Default -WhatIf. ABM/federation deferred. > [CmdletBinding(SupportsShouldProcess=$true)] param([string]$ProvisioningPilotGroup = 'Managed Apple ID Provisioning - Pilot - Stage', [string]$ProvisioningProdGroup = 'Managed Apple ID Provisioning - Corp - Prod', [string]$CodGroup = 'iOS COD - Devices - Pilot', [string]$ByodGroup = 'iOS BYOD - Users - Pilot', [string]$SharedGroup = 'iOS Shared - Devices - Pilot', [string[]]$ProvisioningPilotMembers = @(), [string[]]$ByodMembers = @(), First-party 'Intune Provisioning Client' SP (aka 'Intune Autopilot ConfidentialClient'). For an ASSIGNED Shared device group, this SP must OWN the group or enrollment-time grouping leaves it empty. Matched by AppId, not display name. [string]$ProvisioningClientAppId = 'f1346770-5b25-470b-88bd-d5744ab7952c', [switch]$Apply)

How to run: Preview (default): powershell -ExecutionPolicy Bypass -File .\ABMPROV-Set-ScopeTargeting.ps1 Apply: powershell -ExecutionPolicy Bypass -File .\ABMPROV-Set-ScopeTargeting.ps1 -Apply Add pilot members: ... -Apply -ProvisioningPilotMembers user1@contoso.com,user2@contoso.com Requires the Microsoft Graph PowerShell SDK and delegated scopes: Group.ReadWrite.All, User.Read.All. EXPECTED RESULTS The provisioning (pilot/prod) and per-scenario (COD devices, BYOD users, Shared devices) security groups exist (created if missing); any supplied members are added. Re-run ABMPROV-Report.ps1 to confirm membership. .NOTES Entra ID Solutions — Entra ID + ABM federation & auto-provisioning. Default -WhatIf. ABM/federation deferred. > [CmdletBinding(SupportsShouldProcess=$true)] param([string]$ProvisioningPilotGroup = 'Managed Apple ID Provisioning - Pilot - Stage', [string]$ProvisioningProdGroup = 'Managed Apple ID Provisioning - Corp - Prod', [string]$CodGroup = 'iOS COD - Devices - Pilot', [string]$ByodGroup = 'iOS BYOD - Users - Pilot', [string]$SharedGroup = 'iOS Shared - Devices - Pilot', [string[]]$ProvisioningPilotMembers = @(), [string[]]$ByodMembers = @(), First-party 'Intune Provisioning Client' SP (aka 'Intune Autopilot ConfidentialClient'). For an ASSIGNED Shared device group, this SP must OWN the group or enrollment-time grouping leaves it empty. Matched by AppId, not display name. [string]$ProvisioningClientAppId = 'f1346770-5b25-470b-88bd-d5744ab7952c', [switch]$Apply)

ABMPROV-Validation.ps1

Purpose: For supplied users, reports whether they are in the provisioning scope group and whether the provisioning job is active (so a Managed Apple ID is/should be provisioned and federated). For supplied devices (or all iOS devices), reports the enrollment model so you can confirm each is the intended COD / BYOD / Shared shape. Read-only against Microsoft Graph. The authoritative Managed-Apple-ID state lives in ABM and is deferred to Apple's documentation; this validates the Entra/Intune side that drives it. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\ABMPROV-Validation.ps1 -UserUpns user1@contoso.com powershell -ExecutionPolicy Bypass -File .\ABMPROV-Validation.ps1 -DeviceNames iPhone-COD-01 (No params = validates all managed iOS devices.) Requires the Microsoft Graph PowerShell SDK and delegated scopes: Application.Read.All, Synchronization.Read.All, Group.Read.All, User.Read.All, DeviceManagementManagedDevices.Read.All. EXPECTED RESULTS Per user: in-scope (yes/no) + provisioning job active (yes/no). Per device: enrollment profile type mapped to COD (supervised + user affinity), BYOD (user enrollment), or Shared (no user affinity). .NOTES Entra ID Solutions — Entra ID + ABM federation & auto-provisioning. Read-only. ABM state deferred. > [CmdletBinding()] param([string]$AppDisplayName = 'Apple Business Manager - Federation - Prod', [string[]]$ScopeGroups =

@('Managed Apple ID Provisioning - Pilot - Stage','Managed Apple ID Provisioning - Corp - Prod'), [string[]]$UserUpns = @(), [string[]]$DeviceNames = @())

How to run: powershell -ExecutionPolicy Bypass -File .\ABMPROV-Validation.ps1 -UserUpns user1@contoso.com powershell -ExecutionPolicy Bypass -File .\ABMPROV-Validation.ps1 -DeviceNames iPhone-COD-01 (No params = validates all managed iOS devices.) Requires the Microsoft Graph PowerShell SDK and delegated scopes: Application.Read.All, Synchronization.Read.All, Group.Read.All, User.Read.All, DeviceManagementManagedDevices.Read.All. EXPECTED RESULTS Per user: in-scope (yes/no) + provisioning job active (yes/no). Per device: enrollment profile type mapped to COD (supervised + user affinity), BYOD (user enrollment), or Shared (no user affinity). .NOTES Entra ID Solutions — Entra ID + ABM federation & auto-provisioning. Read-only. ABM state deferred. > [CmdletBinding()] param([string]$AppDisplayName = 'Apple Business Manager - Federation - Prod', [string[]]$ScopeGroups = @('Managed Apple ID Provisioning - Pilot - Stage','Managed Apple ID Provisioning - Corp - Prod'), [string[]]$UserUpns = @(), [string[]]$DeviceNames = @())